Experiment no 4

Name: Afifa Syed **Roll no: 2511139** **Subject: ASD&D**

Aim: To understand Continuous Integration, Jenkins Installation

LO mapped: LO1, LO3

Theory:

Continuous Integration is a development practice that promotes collaboration, early issue detection, and automated testing to enhance the quality and efficiency of software development.

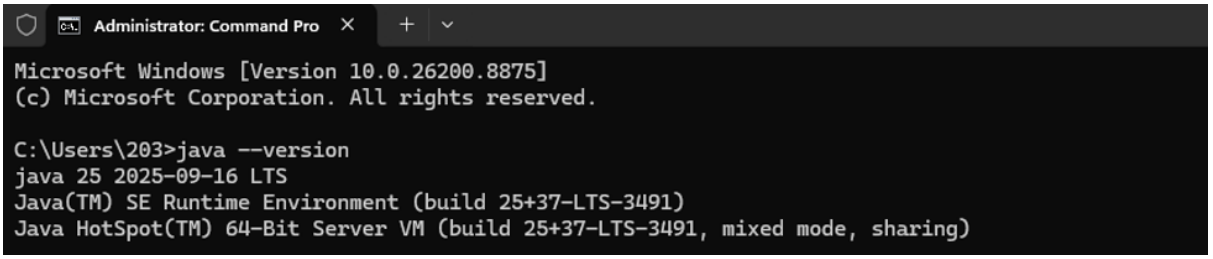
Jenkins is an open-source automation server that facilitates Continuous Integration and Continuous Delivery (CI/CD) of software projects. It helps developers automate various stages of the software development lifecycle, from building and testing to deployment and monitoring. Jenkins supports a wide range of plugins and integrations, making it a versatile tool for automating various tasks in software development.

Steps to be followed while installing Jenkins:

I. Java Installation

Ensure Java Development Kit (JDK) is installed on your system.

Fig. 1



II. Download Jenkins

Get the latest Jenkins WAR file from the official website or package manager.

Fig. 2 Install Generic.war file



III. Launch Terminal

Open a terminal/command prompt on your system.

Command:

java -jar C:\Users\Asus\Downloads\jenkins.war --enable-future-java

Fig. 3 Run the above command

```
C:\Users\Lab203\Desktop>java -jar C:\Users\Lab203\Downloads\jenkins.war --enable-future-java
Running from: C:\Users\Lab203\Downloads\jenkins.war
webroot: C:\Users\Lab203\.jenkins\war
2026-08-07 04:14:11.657+0000 [id=3]     INFO    winstone.Logger#logInternal: Beginning extraction from war file
2026-08-07 04:14:11.694+0000 [id=3]     WARNING o.e.j.ee9.nested.ContextHandler#setContextPath: Empty contextPath
2026-08-07 04:14:11.736+0000 [id=3]     INFO    org.eclipse.jetty.server.Server#doStart: jetty-12.1.8; built: 2026-04-01
T01:09:18.991Z; git: c9cdc9aaa434a3665b8a53b4d1cc3684992da649; jvm 25.0.2+10-LTS-69
2026-08-07 04:14:12.158+0000 [id=3]     INFO    o.e.j.e.w.StandardDescriptorProcessor#visitServlet: NO JSP Support for /
, did not find org.eclipse.jetty.ee9.jsp.JettyJspServlet
2026-08-07 04:14:12.208+0000 [id=3]     INFO    o.e.j.s.DefaultSessionIdManager#doStart: Session workerName=node0
WARNING: A terminally deprecated method in sun.misc.Unsafe has been called
WARNING: sun.misc.Unsafe::objectFieldOffset has been called by com.thoughtworks.xstream.converters.reflection.SunUnsafeR
eflectionProvider (file:/C:/Users/Lab203/.jenkins/war/WEB-INF/lib/xstream-1.4.21.jar)
WARNING: Please consider reporting this to the maintainers of class com.thoughtworks.xstream.converters.reflection.SunUn
safeReflectionProvider
WARNING: sun.misc.Unsafe::objectFieldOffset will be removed in a future release
2026-08-07 04:14:12.498+0000 [id=3]     INFO    hudson.WebAppMain#contextInitialized: Jenkins home directory: C:\Users\L
ab203\.jenkins found at: $user.home/.jenkins
2026-08-07 04:14:12.562+0000 [id=3]     INFO    o.e.j.s.handler.ContextHandler#doStart: Started oeje9n.ContextHandler$Co
reContextHandler@7e8e8651{Jenkins v2.568.2,/,b=file:///C:/Users/Lab203/.jenkins/war/,a=AVAILABLE,h=oeje9n.ContextHandler
$CoreContextHandler$CoreToNestedHandler@49ef32e0{STARTED}}
2026-08-07 04:14:12.583+0000 [id=3]     INFO    o.e.j.server.AbstractConnector#doStart: Started oejs.ServerConnector@3ee
39da0{HTTP/1.1, (http/1.1)}{0.0.0.0:8080}
2026-08-07 04:14:12.583+0000 [id=3]     INFO    org.eclipse.jetty.server.Server#doStart: Started oejs.Server@5d332969{ST
ARTING}[12.1.8,sto=0] @850ms
2026-08-07 04:14:12.584+0000 [id=67]    INFO    winstone.Logger#logInternal: Winstone Servlet Engine running: controlPor
t=disabled
2026-08-07 04:14:12.645+0000 [id=60]    INFO    jenkins.model.Jenkins#<init>: Starting version 2.568.2
```

IV. Navigate to Directory

Use the cd command to navigate to the directory containing the Jenkins WAR file.

During the first execution of Jenkins, an initial administrator password is generated and stored in the following directory:

C:\Users\Lab203\.jenkins\secrets\initialAdminPassword

Fig. 4 Copy the Password and Directory path

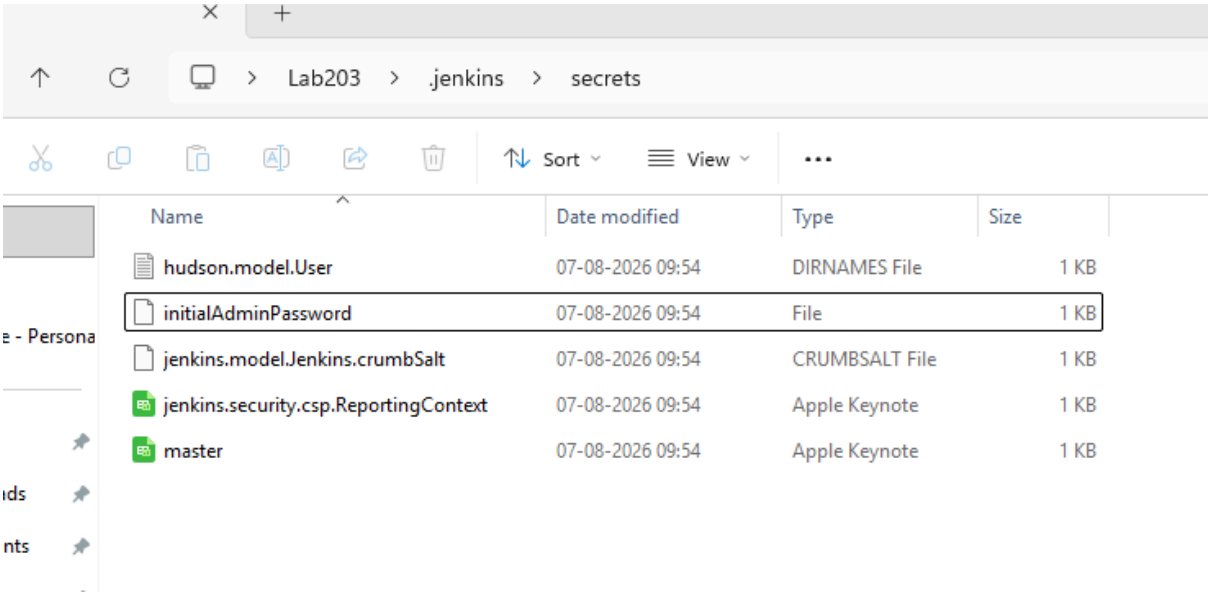


Fig. 5 :

V. Start Jenkins

Run the command:

java -jar jenkins.war

to start the Jenkins server.

```
C:\Users\Lab203\Downloads>java -jar jenkins.war
Running from: C:\Users\Lab203\Downloads\jenkins.war
webroot: C:\Users\Lab203\.jenkins\war
2026-08-07 04:22:03.461+0000 [id=3]     INFO    winstone.Logger#logInternal: Beginning extraction from war file
2026-08-07 04:22:03.495+0000 [id=3]     WARNING o.e.j.ee9.nested.ContextHandler#setContextPath: Empty contextPath
2026-08-07 04:22:03.532+0000 [id=3]     INFO    org.eclipse.jetty.server.Server#doStart: jetty-12.1.8; built: 2026-04-01T01:09:18.991Z; git: c9cdc9aaa
2026-08-07 04:22:03.947+0000 [id=3]     INFO    o.e.j.e.w.StandardDescriptorProcessor#visitServlet: NO JSP Support for /, did not find org.eclipse.jet
2026-08-07 04:22:03.977+0000 [id=3]     INFO    o.e.j.s.DefaultSessionIdManager#doStart: Session workerName=node0
WARNING: A terminally deprecated method in sun.misc.Unsafe has been called
WARNING: sun.misc.Unsafe::objectFieldOffset has been called by com.thoughtworks.xstream.converters.reflection.SunUnsafeReflectionProvider (file:/C:/Us
WARNING: Please consider reporting this to the maintainers of class com.thoughtworks.xstream.converters.reflection.SunUnsafeReflectionProvider
WARNING: sun.misc.Unsafe::objectFieldOffset will be removed in a future release
2026-08-07 04:22:04.266+0000 [id=3]     INFO    hudson.WebAppMain#contextInitialized: Jenkins home directory: C:\Users\Lab203\.jenkins found at: $user
2026-08-07 04:22:04.323+0000 [id=3]     INFO    o.e.j.s.handler.ContextHandler#doStart: Started oeje9n.ContextHandler$CoreContextHandler@7e8e8651{Jenk
AILABLE,h=oeje9n.ContextHandler$CoreContextHandler$CoreToNestedHandler@49ef32e0{STARTED}}
2026-08-07 04:22:04.349+0000 [id=3]     INFO    o.e.j.server.AbstractConnector#doStart: Started oejs.ServerConnector@3ee39da0{HTTP/1.1, (http/1.1)}{0.
2026-08-07 04:22:04.349+0000 [id=3]     INFO    org.eclipse.jetty.server.Server#doStart: Started oejs.Server@5d332969{STARTING}[12.1.8,sto=0] @818ms
2026-08-07 04:22:04.350+0000 [id=67]    INFO    winstone.Logger#logInternal: Winstone Servlet Engine running: controlPort=disabled
2026-08-07 04:22:04.409+0000 [id=60]    INFO    jenkins.model.Jenkins#<init>: Starting version 2.568.2
2026-08-07 04:22:04.462+0000 [id=74]    INFO    jenkins.InitReactorRunner$1#onAttained: Started initialization
2026-08-07 04:22:04.577+0000 [id=86]    INFO    jenkins.InitReactorRunner$1#onAttained: Listed all plugins
2026-08-07 04:22:06.187+0000 [id=110]   INFO    jenkins.InitReactorRunner$1#onAttained: Prepared all plugins
2026-08-07 04:22:06.196+0000 [id=74]    INFO    jenkins.InitReactorRunner$1#onAttained: Started all plugins
2026-08-07 04:22:06.196+0000 [id=91]    INFO    jenkins.InitReactorRunner$1#onAttained: Augmented all extensions
2026-08-07 04:22:06.300+0000 [id=111]   INFO    h.p.b.g.GlobalTimeOutConfiguration#load: global timeout not set
2026-08-07 04:22:06.660+0000 [id=79]    INFO    jenkins.InitReactorRunner$1#onAttained: System config loaded
2026-08-07 04:22:06.660+0000 [id=85]    INFO    jenkins.InitReactorRunner$1#onAttained: System config adapted
2026-08-07 04:22:06.675+0000 [id=86]    INFO    jenkins.InitReactorRunner$1#onAttained: Loaded all jobs
2026-08-07 04:22:06.680+0000 [id=95]    INFO    jenkins.InitReactorRunner$1#onAttained: Configuration for all jobs updated
2026-08-07 04:22:06.704+0000 [id=91]    INFO    jenkins.InitReactorRunner$1#onAttained: Completed initialization
2026-08-07 04:22:06.729+0000 [id=60]    INFO    hudson.lifecycle.Lifecycle#onReady: Jenkins is fully up and running
```

VI. Access Web UI

Open a web browser and go to:

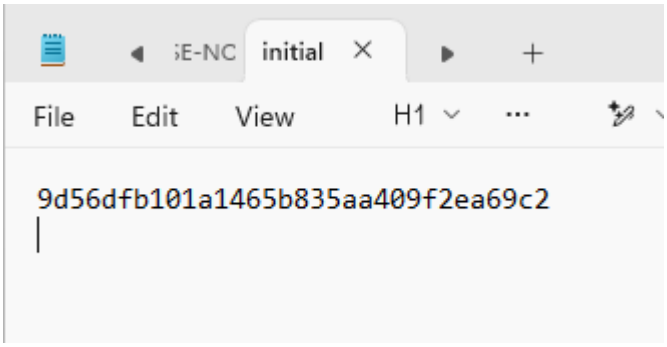
http://localhost:8080

to access the Jenkins setup wizard.

Fig. 6 In browser go to "localhost:8080" and enter your password

VII. Unlock Jenkins

Retrieve the initial admin password from the terminal and enter it in the setup wizard.



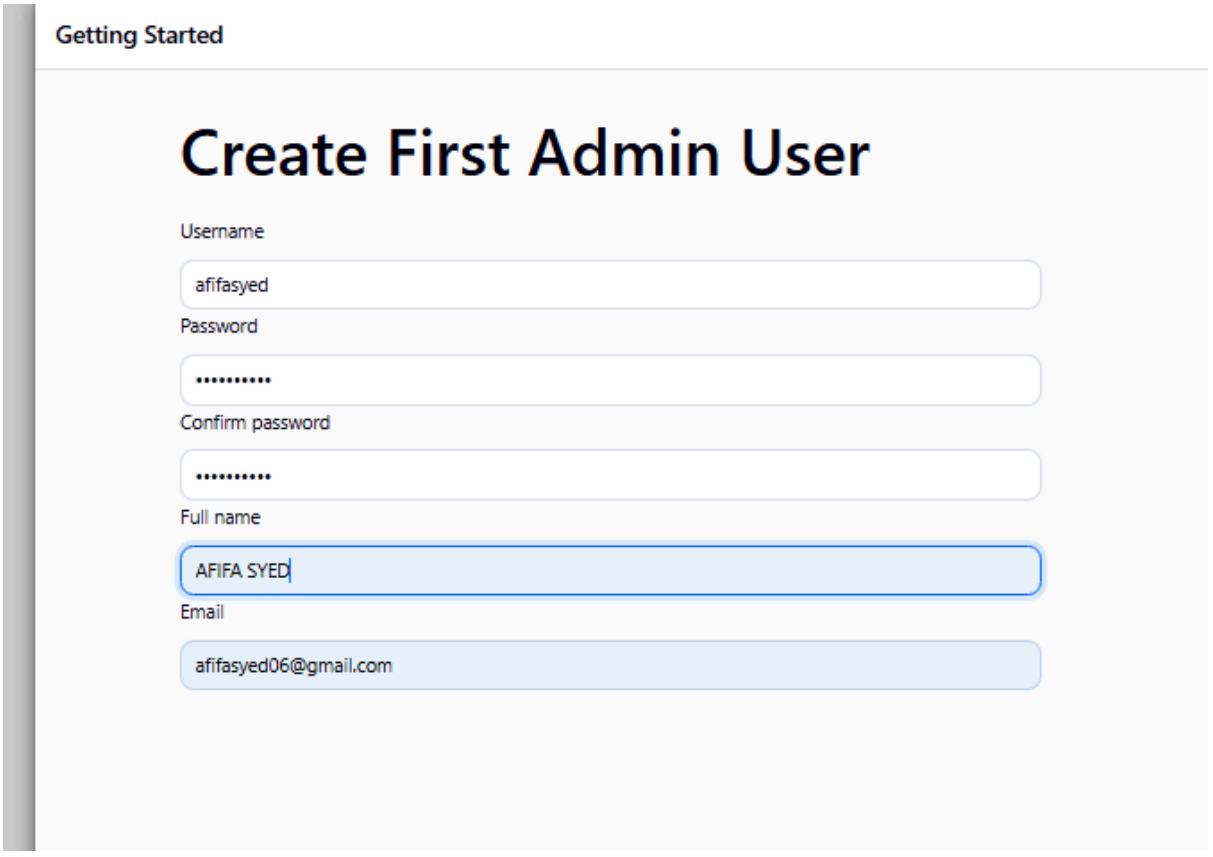
VIII. Customize Plugins

Choose recommended or custom plugins to install during setup.

Fig. 6 After correct password entered you will be redirected to this page. Then click on Install suggested plugins.

IX. Create Admin User

Set up an admin user account with credentials.



X. Jenkins URL

Confirm the Jenkins URL and complete the setup process.

XI. Dashboard

Log in with the admin credentials to access the Jenkins dashboard.

Fig. 7 After successful setup of plugins and Jenkins

Fig. 8 For login you can now use your username and password you gave while setup of Jenkins

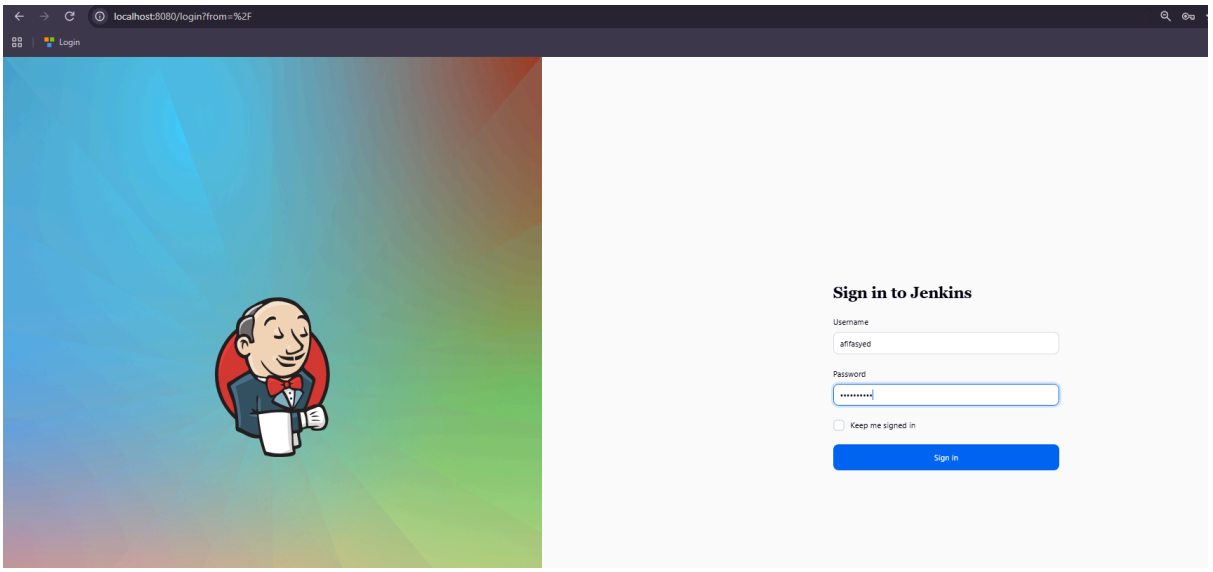
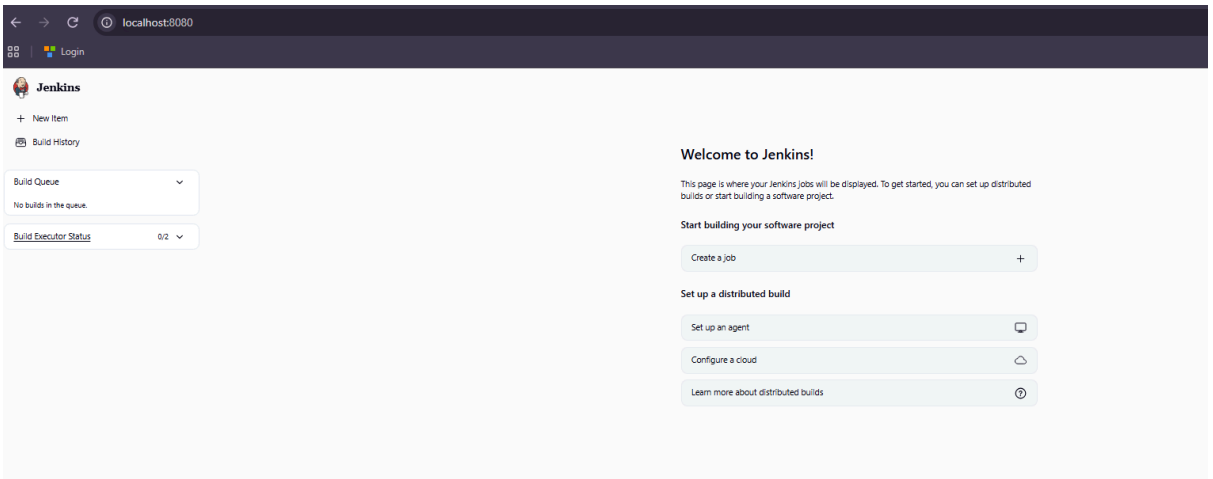


Fig. 9 Dashboard of Jenkins



Conclusion

By this assignment we learned about Continuous Integration and the way to install Jenkins.

AIM: To build Pipeline using Jenkins

THEORY:

What is a Pipeline in Jenkins?

A Jenkins Pipeline is a set of automated processes that help developers and DevOps teams build, test, and deploy code in a Continuous Integration/Continuous Delivery (CI/CD) environment.

In simple terms, it is a workflow that defines how software moves from code to production in an automated and repeatable manner.

Key Benefits of Jenkins Pipeline

- Automates CI/CD processes
- Provides better visibility into the build process
- Offers version control of build logic (Pipeline as Code)
- Supports complex workflows (parallel execution, conditional logic, etc.)

Types of Jenkins Pipelines

There are two main types of syntax used to define Jenkins pipelines.

1. Declarative Pipeline

- Simpler and more structured syntax
- Designed for users who want an easy-to-read format
- Introduced later to make pipelines more readable and maintainable

Key Features

- Uses pipeline {} block
- Organized into stages and steps
- Easier error handling
- Better validation in Jenkins UI

2. Scripted Pipeline

- More flexible and powerful
- Written in Groovy-based syntax
- Gives full control over the build process
- Suitable for advanced users

Key Features

- Uses node {} block
- Everything is written as code
- Allows complex scripting and logic
- Less user-friendly for beginners

1. Declarative Pipeline

CREATION

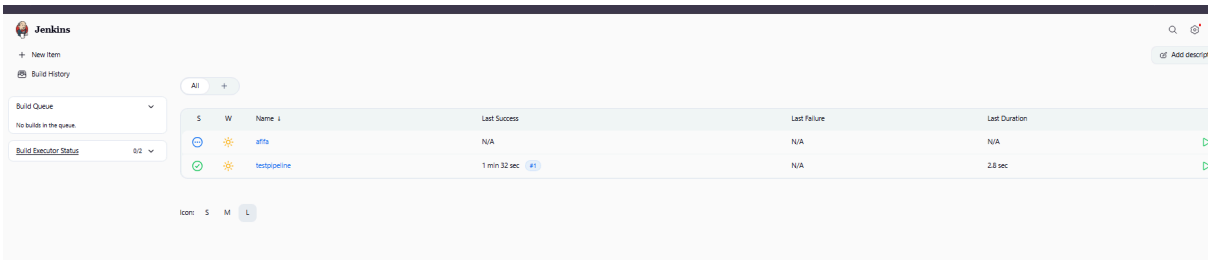
Create a **New Item** in Jenkins and select **Pipeline**.

CODE

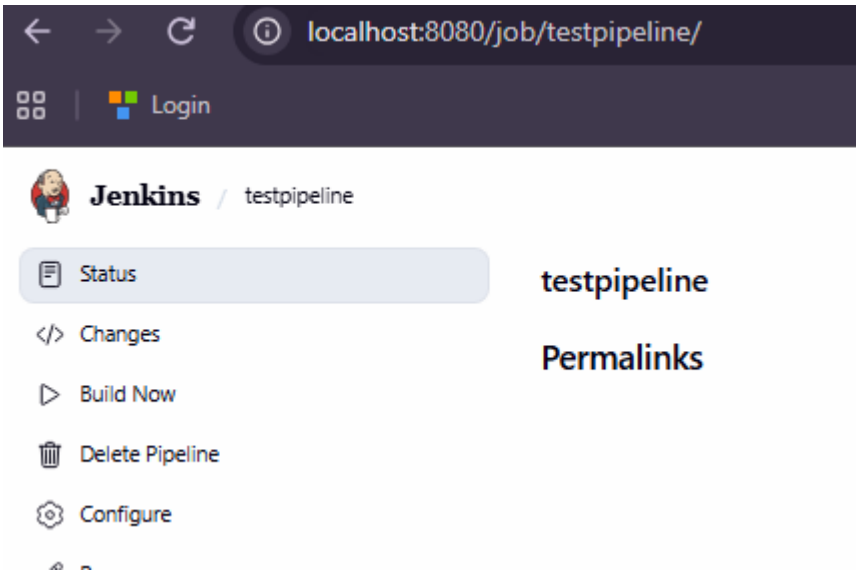
```
pipeline {
    agent any

    stages {
        stage('Hello') {
            steps {
                echo 'Hello, Jenkins!'
            }
        }
    }
}
```

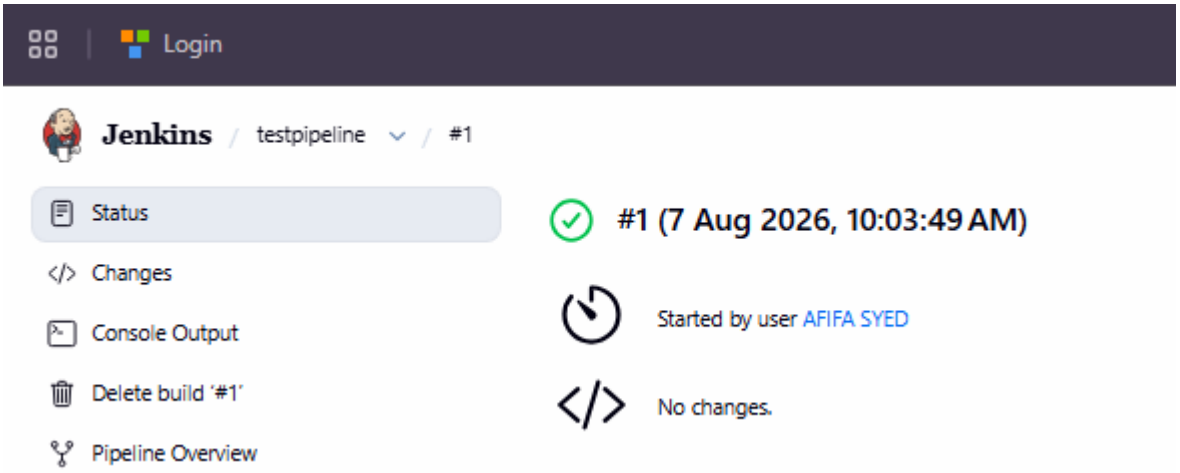
OUTPUT



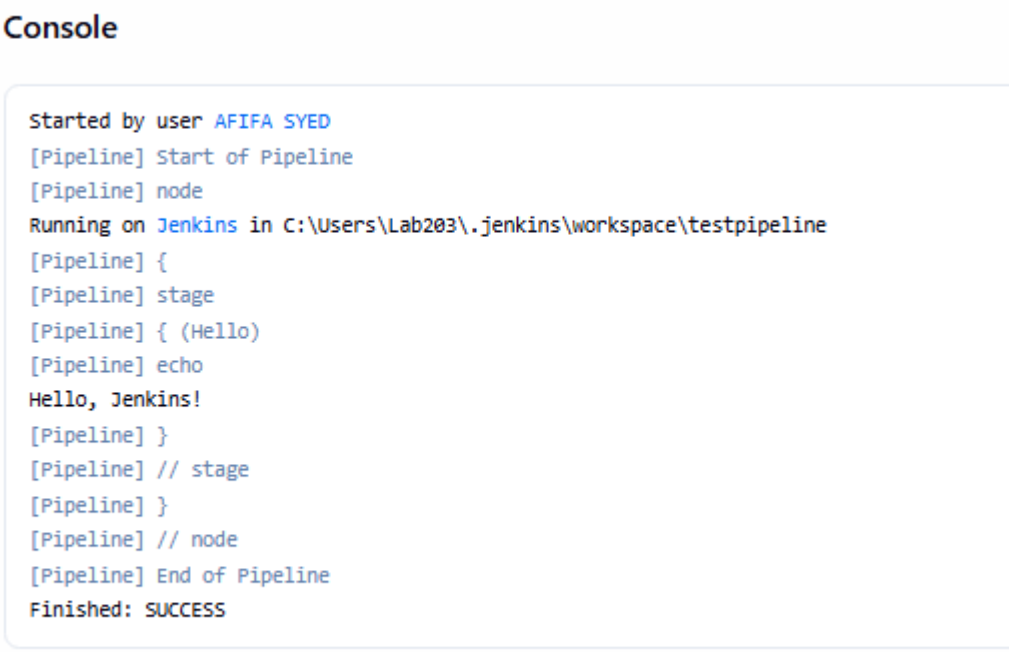
- Create the pipeline.



- Click **Build Now**.



- Open **Console Output**.



Expected output:

Started by user admin

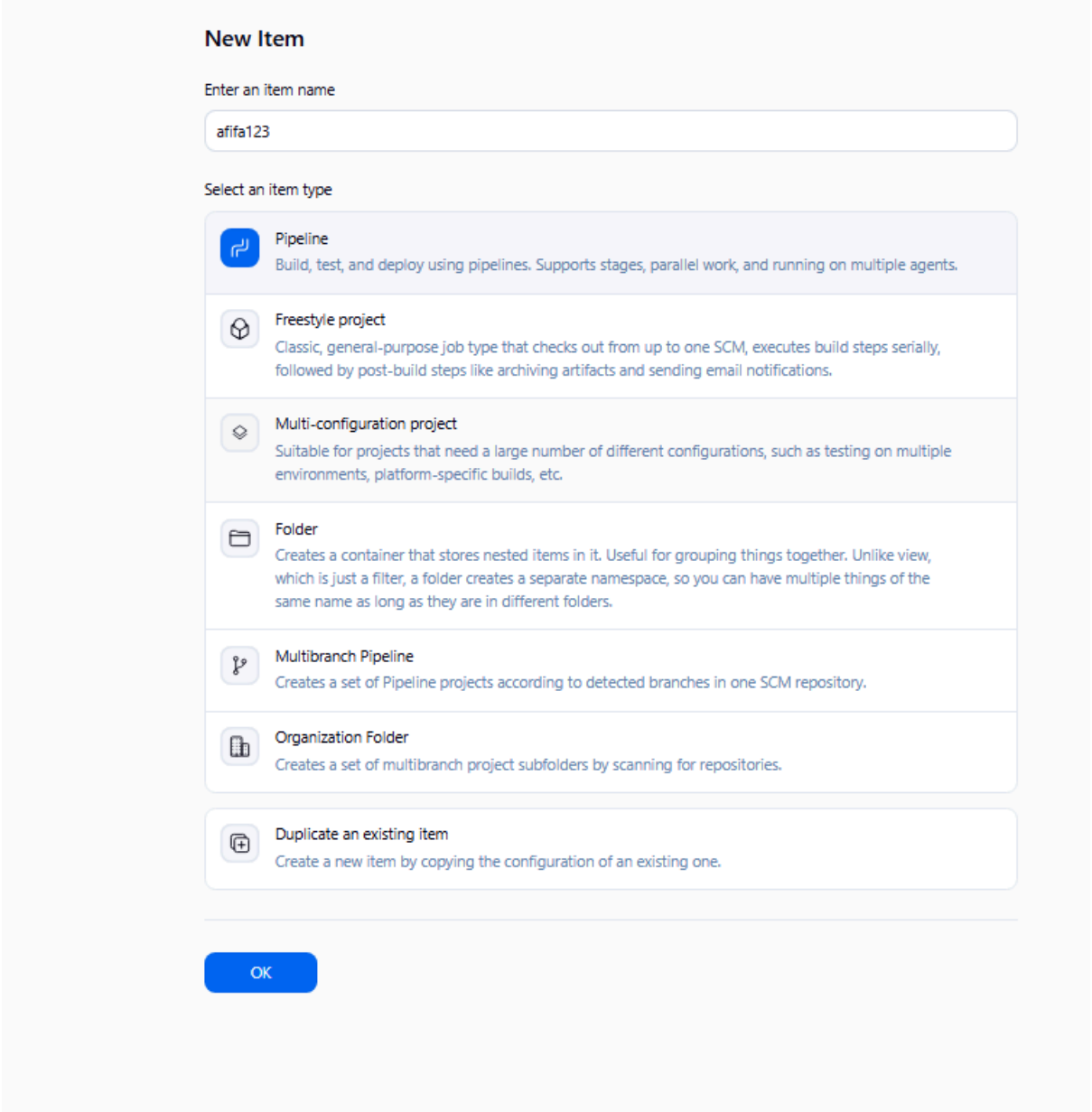
[Pipeline] Start of Pipeline
[Pipeline] node
Running on Jenkins
[Pipeline] stage
[Pipeline] { (Hello)
[Pipeline] echo
Hello, Jenkins!
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline

Finished: SUCCESS

2. Scripted Pipeline

CREATION

1. Click **New Item**.
2. Enter an item name (Example: pipeline2).
3. Select **Pipeline**.
4. Click **OK**.



CONCLUSION (LO1, LO3)

In this assignment, we learned how to create Jenkins Pipelines to automate the build, test, and deployment process. We explored both Declarative and Scripted Pipelines, understanding their structure, benefits, and use cases. By implementing different pipeline examples, we demonstrated how Jenkins enables efficient, repeatable, and reliable CI/CD workflows.